Practical 9 - Yacc/Bison Code

Yacc file: practcc9.y (lines 1-33)

Open practcc9.y

```
/* Name:- Mohammed Zaid Shaikh
Prn:- 24070521043
Sec:- A2
*/



%{
#include <stdio.h>
#include <stdlib.h>

int yylex();
void yyerror(char *);

int power(int base, int exp)
{
    int result = 1;
    int i;

    for(i = 0; i < exp; i++)
        result = result * base;

    return result;
}
%}

%token NUMBER INC DEC MOD

%left '+' '-'
%left '*' '/' MOD
%right '^'

%%
```

Yacc file: practcc9.y (lines 34-72)

```

stmt: expr '\n'    { printf("Result: %d\n", $1); }
| error '\n'   { printf("Invalid expression\n"); yyerrok; }
    ;

expr: NUMBER       { $$ = $1; }
| expr '+' expr { $$ = $1 + $3; }
| expr '-' expr { $$ = $1 - $3; }
| expr '*' expr { $$ = $1 * $3; }
| expr '/' expr { if($3 == 0)
                        yyerror("Division by zero");
                    else
                        $$ = $1 / $3;
                }
| expr MOD expr { if($3 == 0)
                        yyerror("Modulo by zero");
                    else
                        $$ = $1 % $3;
                }
| expr '^' expr { $$ = power($1, $3); }
| '(' expr ')'  { $$ = $2; }
| INC expr      { $$ = $2 + 1; }
| DEC expr      { $$ = $2 - 1; }
| expr INC      { $$ = $1 + 1; }
| expr DEC      { $$ = $1 - 1; }
;

%%

void yyerror(char *s)
{
    printf("Error: %s\n", s);
}

int main()
{
    yyparse();
    return 0;
}
```

Practical 9 - Lex Code and Output

Lex file: practcc9.l

Open practcc9.l

```
/* name:- Mohammed Zaid Shaikh
prn:- 24070521043
sec:- A2
*/


%{
#include <stdio.h>
#include <stdlib.h>
#include "y.tab.h"
%}

%%

[0-9]+      { yylval = atoi(yytext); return NUMBER; }
"++"        { return INC; }
"--"        { return DEC; }
"%"         { return MOD; }
[+\-*/^]  { return yytext[0]; }
[ \t]       ;
\n          { return '\n'; }
.           { return yytext[0]; }

%%

int yywrap()
{
    return 1;
}
```

Output

```
zaid787@LAPTOP-ID0Q783M:~$ gedit practcc9.l
zaid787@LAPTOP-ID0Q783M:~$ gedit practcc9.y
zaid787@LAPTOP-ID0Q783M:~$ yacc -d practcc9.y
practcc9.y: warning: 28 shift/reduce conflicts [-Wconflicts-sr]
practcc9.y: note: rerun with option '-Wcounterexamples' to generate conflict counterexamples
zaid787@LAPTOP-ID0Q783M:~$ flex practcc9.l
zaid787@LAPTOP-ID0Q783M:~$ gcc y.tab.c lex.yy.c -o a.out
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
5 + 25 * 3
Result: 80
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
45 % 7 - 2
Result: 1
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
--9
Result: 8
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
++10
Result: 11
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
3 ^ 5
Result: 243
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
10++
Result: 11
zaid787@LAPTOP-ID0Q783M:~$ 10+
10+: command not found
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
10+
Error: syntax error
Invalid expression
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
0/9
Result: 0
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
9 / 0
Error: Division by zero
Result: 9
zaid787@LAPTOP-ID0Q783M:~$
```